



TILT WALL BUILDING DESIGN

PURPOSE

This document establishes guidelines and recommended procedures for the design of field cast tiltup concrete building wall panels. Such guidelines and procedures are limited to exterior, nonblast resistant, and load bearing panels, and nonload bearing panels.

SCOPE

This document includes the following major sections:

- DESIGN CONDITIONS
- ROOF AND FLOOR FRAMING
- DESIGN CONSIDERATIONS
- WALL DESIGN
- REFERENCES
- ATTACHMENTS
- APPLICATION

This document may be used for the design of precast tilt wall buildings typically found in commercial and industrial facilities.

DESIGN CONDITIONS

Concrete tiltup structures must be designed to resist a wide variety of loading conditions, both vertical and horizontal. The concrete panels may be designed as bearing or nonbearing, shearwalls or nonshearwalls. For each of these cases, proper loading conditions must be considered and proper detailing must be done to ensure that elements perform as they are intended. As an example, walls that are intended to be strictly partition walls must be detailed to eliminate possible bearing and diaphragm load transfer. Conversely, walls designed as bearing shearwalls must be detailed to properly develop all of the design loads.

Design Criteria

Design loads should follow requirements of the applicable code of the governing jurisdiction, or of project specification. As an example, sample references to the UBC (Uniform Building Code) have been used in this document.

Vertical Loads

Vertical loads consist of dead loads and/or live loads. These loads result in both an axial component and a flexural component. Flexural stresses are caused by eccentricity of the floor connection as well as P-delta effects.



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DEAD

Dead loads may include the following:

- Diaphragm (plywood or decking)
- Subframing
- Beams and girders
- Mechanical and electrical equipment
- Roofing materials
- Ceiling assemblies (if any)
- Roof mounted equipment
- Wall weight

LIVE

Live loads are as specified in the applicable codes) or project specific criteria.

SNOW

Snow loads will be designed as required by local criteria.

WIND

Vertical effects of wind loads should be considered, particularly for lightweight roof systems where suction may create net upward loads.

Load Combinations

Load Combinations shall be calculated in accordance with Structural Engineering Specification 000.215.00910: Structural Engineering Criteria.

Horizontal Loads

Horizontal loads generally include some or all of the following:

- Seismic
- Wind
- Snow drift
- Active earth pressure
- Blast (not covered)
- Internal explosion containment (not covered)

Out of Plane Loads

Out of plane loads are those which act perpendicular to the plane of the wall. These loads are distributed to the horizontal floor elements through beam action of the wall. These forces may be caused by the following:



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- Seismic
- Wind
- Snow drift
- Lateral earth pressure
- Blast (not covered)
- Internal explosion containment (not covered)

Inplane Loads

Inplane loads are distributed by the floor diaphragms to the shearwalls.

Lifting Loads

Lifting (gravity) loads on the panel during erection are potentially the largest load the panel will resist during its lifetime. It is often necessary to provide intermediate pickup points and local temporary strengthening (strongbacks or reinforcing steel) to reduce stresses.

For many years, it was the responsibility of the Design Engineer to include this in the design. With the introduction of computers and specialty concrete accessories, the design responsibility has shifted to the supplier. The following points summarize the steps and issues to consider when designing for erection:

- Determine the CG (Center of Gravity) of panel, both horizontal and vertical. It is important that the panel hang during lifting. Additionally, the pickup points should be located above the CG to ensure the lower edge will be let down first.
- Lifting inserts at the pickup points should be set in pairs so that a single cable and sheave (pulley) can distribute loads evenly to each point.
- Bending stresses between pickup points should be checked for tension in concrete, with a modulus of rupture of $7.5\sqrt{f'_c}$. During lifting, the compressive strength should be checked with cylinder breaks. A minimum strength of 2,500 psi is usually specified. Panels are designed as strips of slab spanning between pickup points.
- Additional reinforcing may be added to assist the panel in spanning between lift points. This reinforcing is specified by the supplier on shop drawings. In lieu of additional reinforcing, the supplier may elect to reinforce the panel using strongbacks which may be reused on several different panels.
- The inserts are checked for pullout using the maximum calculated lift load. An appropriate factor of safety (2.5 to 4) is used to allow for unforeseen loads (panel jerking and panel sticking). The inserts should also be checked for shear as the panel transitions from horizontal to vertical.



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ROOF AND FLOOR FRAMING

Refer to Structural Engineering Guideline 000.215.1244: Masonry Building Design, for a description of typical framing systems.

DESIGN CONSIDERATIONS

Certain layout and construction considerations that are unique to tiltup buildings necessitate coordination and interface of the engineer and architect.

Building Layout

The building layout depends upon the type of framing system used. Panel joints and pilaster spacing must be coordinated to accommodate girder spacing. Panelized roofs, precast prestressed double tees, and other specialized systems all have standard spacing which would effect panel layouts.

Wall panel sizes must be coordinated with the architect to satisfy building geometry. Panel joints are often limited by door and window locations, joint symmetry, or architectural reveals. The maximum size of panels is also limited by crane capacity, casting bed limitations, or other site specific physical constraints such as access roads.

Construction Sequence

The following is a basic outline of the tiltup construction process. Many jobs require variations to this procedure, but follow the same basic philosophy.

OUTLINE

- Pour casting bed (or building floor slab, if utilized for casting). If the building slab is utilized for casting, building footings and wall footings should be in place; otherwise, they may be poured concurrently.
- Lay out wall panels on the casting surface and construct perimeter forms. At this time, any insert face treatments, reveal strips, or other embedded items are set in place. The bottom face is commonly used as the exterior face.
- Doors and windows are blocked out and the reinforcing steel is set in place.
- Concrete is then set in place, finished, and allowed to cure for 7 to 10 days minimum.
- Once walls have reached the required lifting strength, the predetermined pick up points are utilized to lift the panel into place. The lifting sequence should be



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predetermined to allow a logical erection of panels around the building while not restricting erection of interior panels.

- Panels are aligned and leveled; temporary bracing is secured until final attachment to the roof is made.
- Panels are connected or caulked at the joints. Blemishes and cracks are cleaned and repaired.
- Pourstrip at the building floor slab is set to tie the wall base in place.
- Once the final roof or intermediate floor levels are constructed, the braces are removed.

Foundation Types

There are 2 basic types of foundation systems utilized in tiltup wall design:

Continuous Footing

The first type of system is to rest the panel on a continuous footing along the length of the panel. The wall is connected to the floor slab in the pour strip which provides support for out of plane loading. The panel is connected to the footing through dowels or by embedded angles or channels which are then welded after the panel is in place. This connection is designed to transfer in place shear loads. Panel overturning (or net uplift) is controlled by hold down connections located as close to the panel ends as possible. The footing itself is sized to carry the vertical loading.

Spread Footings

The second system is the isolated (or spread) footing approach. Spread footings are located at panel ends (also generally where pilasters occur) to support the weight of the panel. The panels are tied to the floor slab similar to the previous method. The inplane load and net uplift must both be designed to transfer into the spread footings at the ends. The footing itself is sized to carry the vertical loading, but must be checked for eccentric loading during erection, or for support of panels of different size (or tributary floor loading). For eccentric loading during construction, a 100 percent increase of allowable soil bearing values is often acceptable. Such increases should be verified by a geotechnical engineer prior to design.

Joint Connections

Joints between panels can either be open (and filled with expansion filler) or closed by cast in place pourstrip elements. Each method has its strengths and weaknesses.



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Open Joint

- No in place forming required
- Allows for panel shrinkage
- Pilasters are formed on the ground with the panel
- Easier to connect
- Requires fire rated fillers at fire rated walls

Poured In Place Joints

- Provides a continuous wall
- Column provides lateral support to both panel edges
- Simplifies panel forming
- More difficult for pilaster forming
- Joints are fire rated
- No chord welding required

Chord Bars

Since the building design relies upon diaphragm action of the roof or floors, chord elements must be designed to resist the tension and compression components.

Chord bars or chord reinforcing is often added within the wall panel, following the roof line to develop these forces. An alternative to these bars is to utilize a continuous steel ledge with a weld splice at the panel joints.

Chord reinforcing for closed joint wall systems is spliced at and developed through the cast in place pilaster element. For open joint wall systems, the cord bars are usually welded to a special splice pocket using short pieces of angle as the splice. To control cracking, the chord bars within the panel are wrapped in plastic sleeves which allow the panel to shrink inward without cracking the panel.

Drag Struts

Drag strut connections are utilized to transfer inplane loads from the diaphragm to the wall panel. The load is generally transferred from the diaphragm to the ledger and then into the wall. The following components must be checked:

- Diaphragm to ledger transfer, such as welding or nailing
- Adequacy of bolts in ledger
- Adequacy of drag bars (the size of bars is often governed by the chord forces)
- Any special plates or welds as detailed



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Hold Downs

There are many types of hold down details. There are, however, certain basic considerations to keep in mind:

- Detail must develop entire uplift force from the wall panel through connection and into the footing.
- Detail must allow for field tolerance adjustment (horizontal and vertical).
- Detail must allow for clearance around building columns or other restrictive elements.
- Detail must maximize shop labor and minimize field labor.

WALL DESIGN

The primary concern for the design of tall slender wall panels is lateral instability and buckling due to lateral loads. However, other critical forces to consider are vertical (dead and live) loads, as well as the eccentric effects of the loads caused by the P-delta effect. Inplane shear loads and effects of uplift are also of major concern.

This section will cover the main criteria in tiltup wall design.

Minimum Thickness (Panel Walls)

An absolute minimum panel thickness is 5- 1/2 inches. The recommended minimum thickness is 6 inches. Panel thicknesses should, if possible, match standard wood form sizes, such as 6- 1/2 inches or 7- 1/4 inches.

Minimum Reinforcing

A minimum area of wall reinforcing will be provided as follows:

Vertical

A minimum ratio of reinforcement to gross area will be:

(Refer to ACI 318, Section 14.3.2.)

- 0.0012 for No. 5 and smaller, F_y 60 ksi
- 0.0015 for all other bars
- 0.0020 for welded wire fabric



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Horizontal

- A minimum ratio of reinforcement to gross area will be:
- (Refer to ACI 318, Section 14.3.3.)
- 0.002 for No. 5 and smaller, Fy 60 ksi
- 0.0025 for all other bars
- 0.0020 for welded wire fabric

Other Criteria

- Walls greater than 10 inches shall have 2 layers of reinforcing
- Reinforcing shall not be spaced more than 3 times the wall thickness, nor 18 inches
- Provide two No. 5 bars around all door and window openings, extended at least 24 inches beyond openings.

Empirical Design Method

Walls in which the resultant of all factored loads are located within the middle third of the overall wall thickness may be designed using the criteria of this section. Walls not conforming to this criteria must be designed for slenderness effects. Refer to UBC, all, Section 1914.5, Empirical Design Method. Alternately, walls could be designed using methods outlined in ACI 318, Section 14.4.

Minimum Thickness (Bearing Walls)

The thickness of bearing walls will be no less than 1/25 of the unsupported height or length, whichever is shorter, or

$$\frac{l}{t} \leq 25$$

Design Axial Strength

The design axial strength P_{nw} will be computed by

$$\phi P_{nw} = 0.55 \phi f' A_g \left[1 - \left(\frac{klc}{32h} \right)^2 \right]$$

where

$k = 0.7$ Effective length factor

$k = 0.8$ Fixed against rotation 1 or both ends



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$k = 1.0$ Pinned both ends
 $k = 2.0$ Walls not braced against translation
 A_g = Gross area
 f'_c = Specified compressive strength
 h = Overall thickness
 l_c = Distance between supports
 $\phi = 0.7$ Strength Reduction Factor

Slender Wall Design

The following section provides alternate design criteria which satisfies the slenderness effects criteria of UBC, Section 1910.10.

Limitations

- Vertical service load stress at the location of maximum moment does not exceed $0.04 f'_c$.
- Reinforcement ratio ρ does not exceed $0.6 \rho_b$, where ρ_b is the ratio that produces the balanced strain condition.
- Sufficient reinforcement is provided so that the nominal moment capacity times the ϕ factor is greater than M_{cr}
- Distribution of concentrated load does not exceed the width of bearing plus a width increasing at a slope of 2 vertical to 1 horizontal down to the design flexural section.

Design Strength

Required factored moment, M_u , at the midheight cross section for combined axial and lateral factored loads, including the P-Delta moments, will be limited by

$$M_u \leq \phi M_n$$

Deflection

The P-delta moments will be determined using the maximum potential deflection, Δ_n :

$$\Delta_n = \frac{5N_n l_c^2}{48E_c I_{cr}}$$

The midheight deflection, Δ_s , under unfactored service loads, will be limited to:



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$$\Delta_s \leq \frac{l_c}{150}$$

where

$$\Delta_s = \Delta_{cr} + \left(\frac{M_s - M_{cr}}{M_n - M_{cr}} \right) (\Delta_n - \Delta_{cr})$$

$$\Delta_{cr} = \frac{5M_{cr}l_c^2}{48E_cI_g}$$

$$M_{cr} = \frac{5\sqrt{f'_c I_g}}{y_t}$$

$$I_{cr} = nA_{se}(d - c)^2 + \frac{bc^3}{3}$$

$$A_{se} = \frac{p_u + A_s f_y}{f_y}$$

where

M_s = Unfactored service load moment

M_{cr} = Cracked moment

M_n = Nominal moment strength

E_c = Modulus of elasticity of concrete

l_c = Vertical distance between supports

I_{cr} = Moment of Inertia of cracked section

I_g = Moment of Inertia of gross section

y_t = Distance from centroid to extreme fiber (neglecting reinforcement)

Δ_{cr} = Cracked deflection

A_{se} = Effective area of reinforcement

p_u = Factored axial load

A_s = Area of reinforcement

REFERENCES

UBC (Uniform Building Code), 1997 Edition

ACI (American Concrete Institute) 318-99

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Connections for Tilt-up Wall Construction. Portland Cement Association. 1987

Design Handbook For Precast and Prestressed Concrete. PCI (Prestressed Concrete Institute). Fourth Edition 1992

Tilt-up Construction. ACI (American Concrete Institute). Compilation No. 4, April 1980.

Structural Engineering

Specification: 000.215.00910: Structural Engineering Criteria

Structural Engineering

Guideline: 000.215.1215: Wind Load Calculation

Structural Engineering

Guideline: 000.215.1216: Earthquake Engineering

Structural Engineering

Guideline: 000.215.1244: Masonry Building Design

ATTACHMENTS**Attachment 01**

Typical Panel Hold-Down Detail

Attachment 02

Typical Wall/Footing Details at Exterior Details

Attachment 03

Typical Wall/Footing Details

Attachment 04

Typical Panel Joint Details

Attachment 05

Typical Pilaster Detail

Attachment 06

Direct Bearing Connections

Attachment 07

Steel Or Concrete Haunch Bearing Connections

Attachment 08

Angle Seat Bearing Connections

Attachment 09

Welded Plate Tieback Connections



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Attachment 10

Welded Angle Tieback Connections

Attachment 11

Threaded Rod Tieback Connections

Attachment 12

Welded Alignment Connections

Note: *Calculations for Attachment 13 through 15 have not been updated to meet the requirements of the latest references and design codes.*

Attachment 13

Sample Design 1: Solid Panel on Continuous Footing

Attachment 14

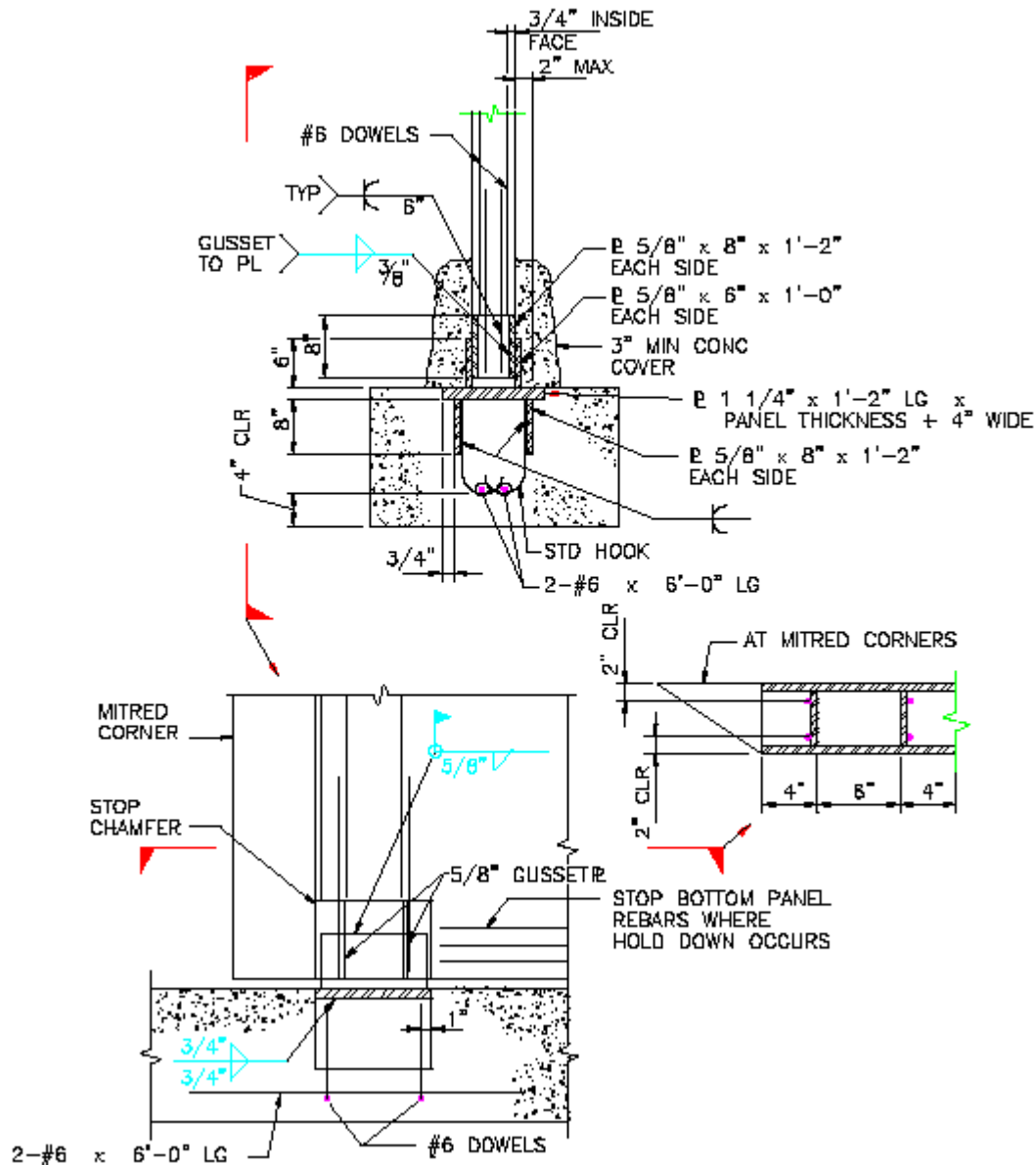
Sample Design 2: Wall Panel with Opening

Attachment 15

Sample Design 3: Beam/Panel Connection

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Typical Panel Hold-Down Detail



TYPICAL WALL/FOOTING DETAILS AT EXTERIOR DETAILS

Figure 1. Continuous Footing

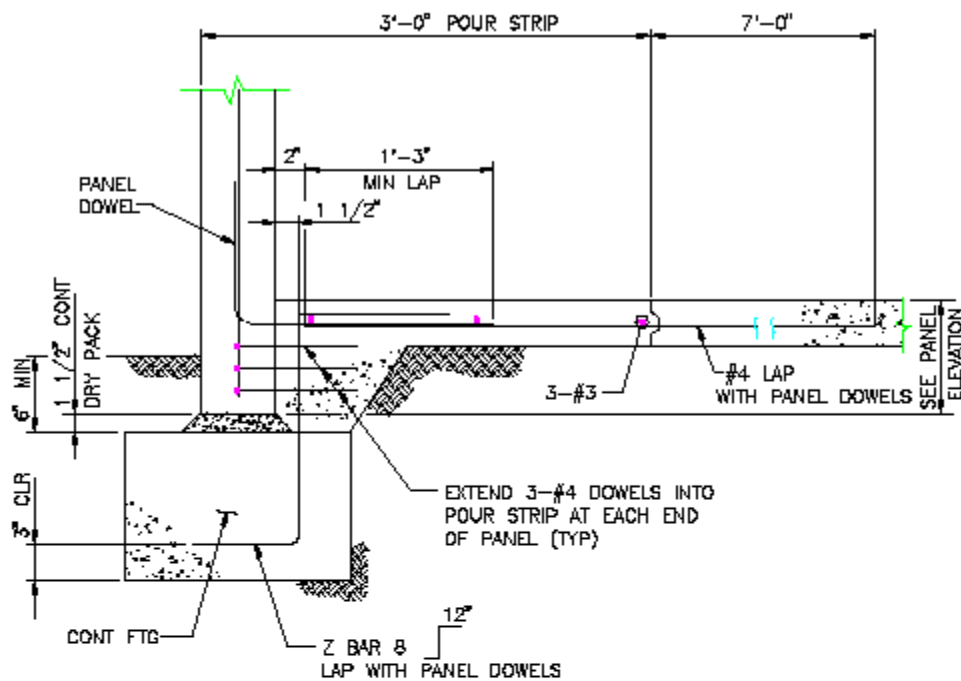
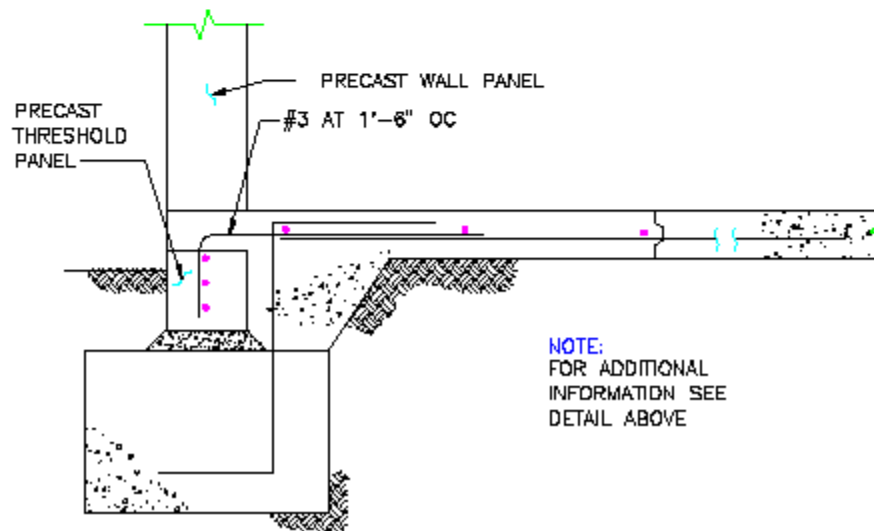


Figure 2. Continuous Footing at Opening



TYPICAL WALL/FOOTING DETAILS AT EXTERIOR DETAILS

Figure 1. Typical Wall/Footing Details at Interior Footing

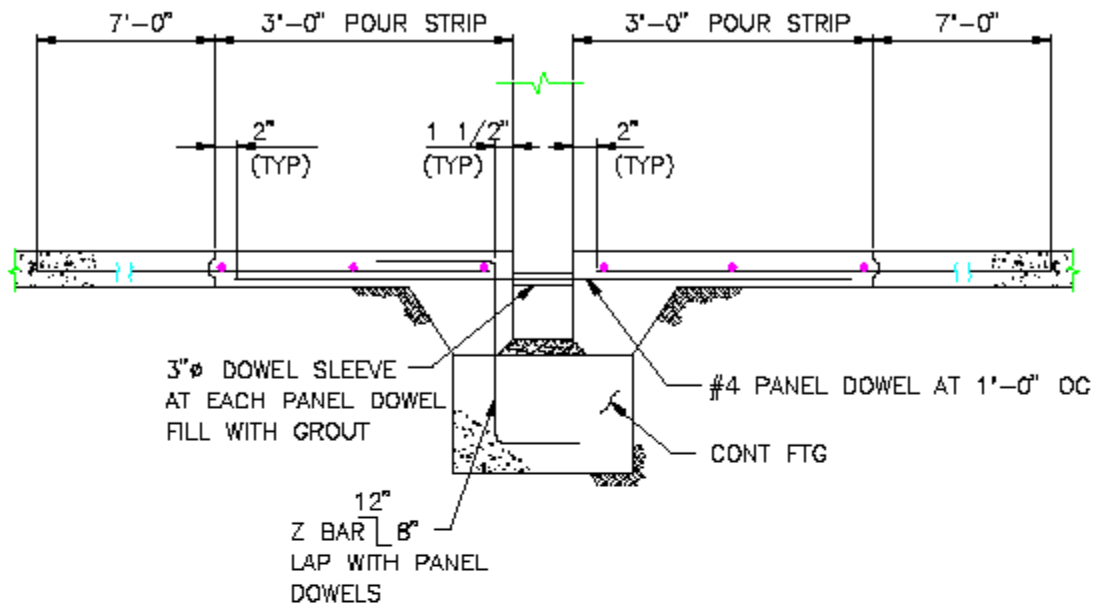
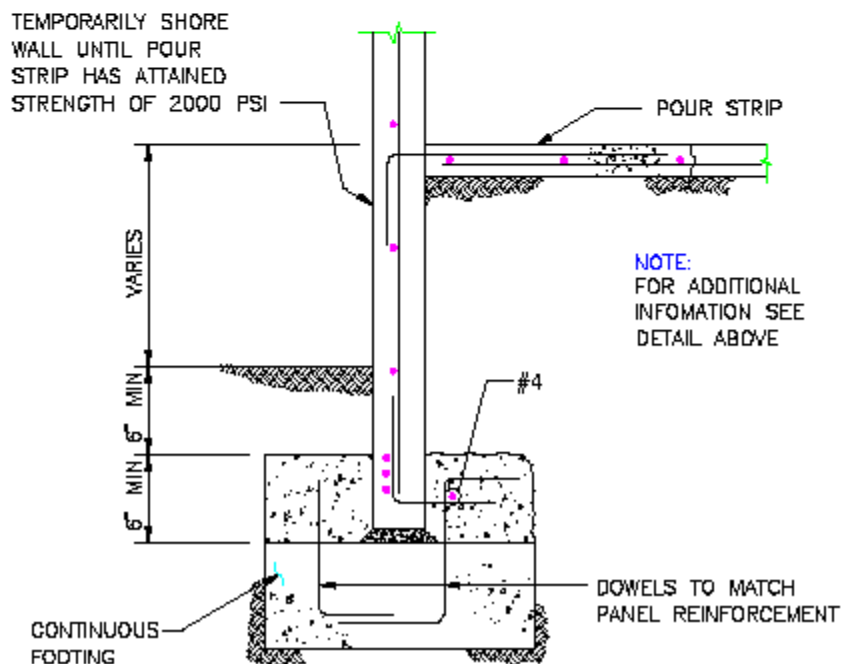
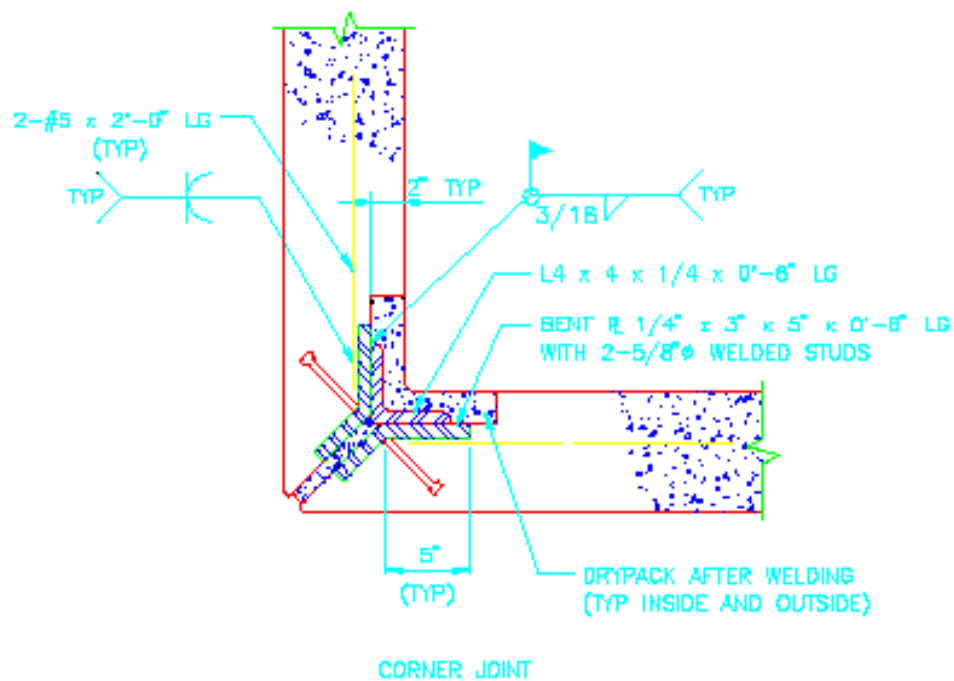
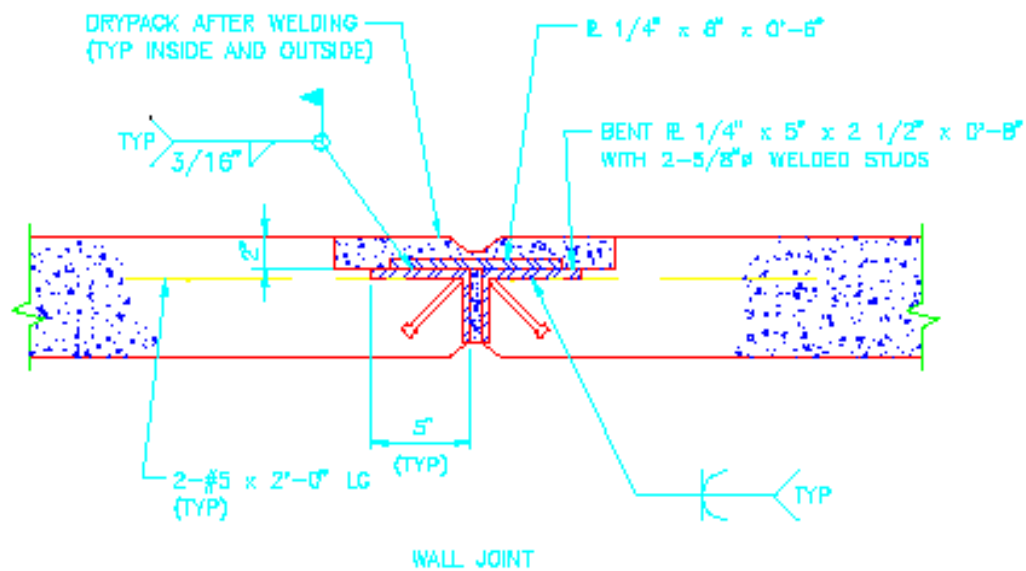


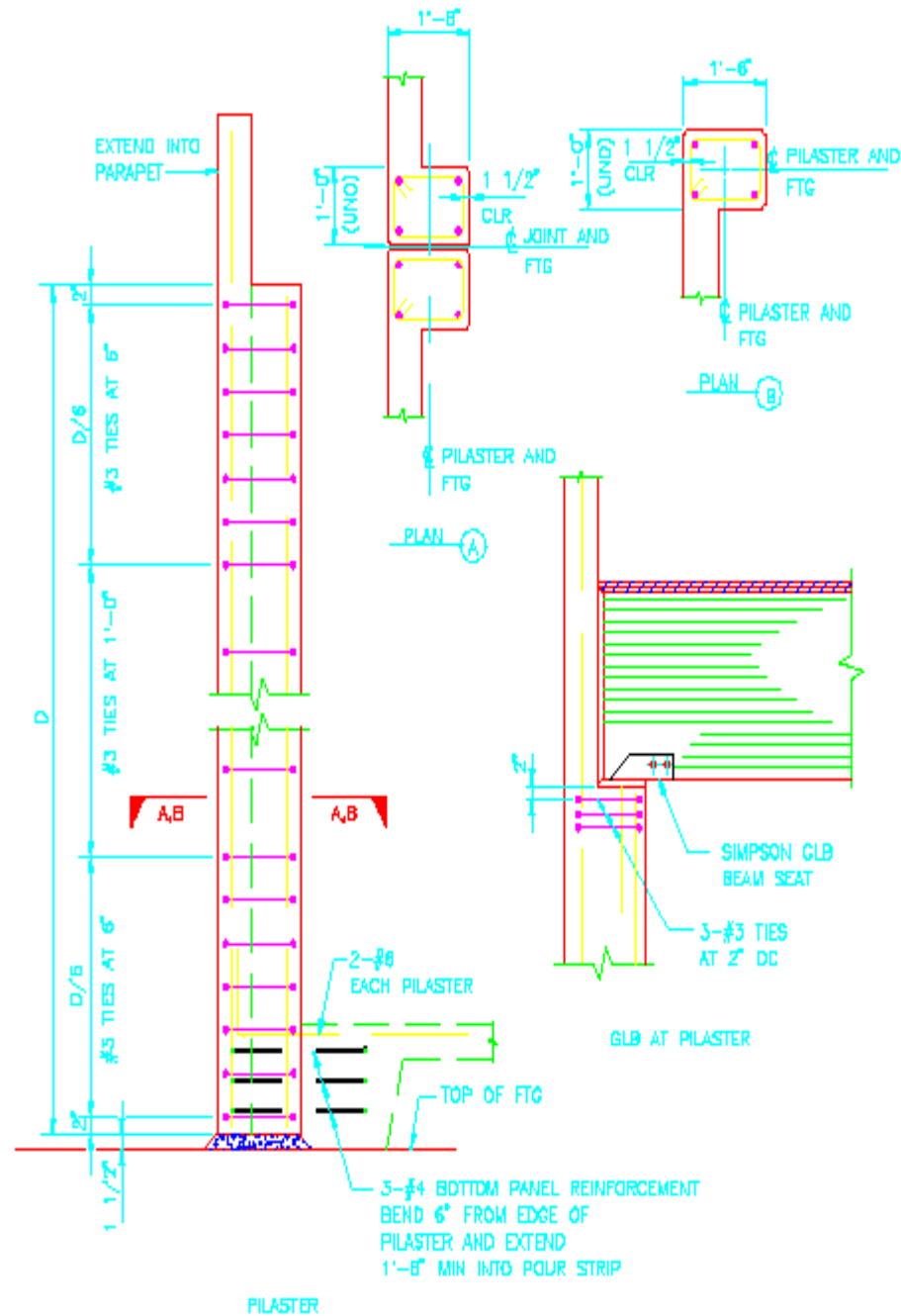
Figure 2. Typical Wall/Footing Details at Footing with Elevated Slab



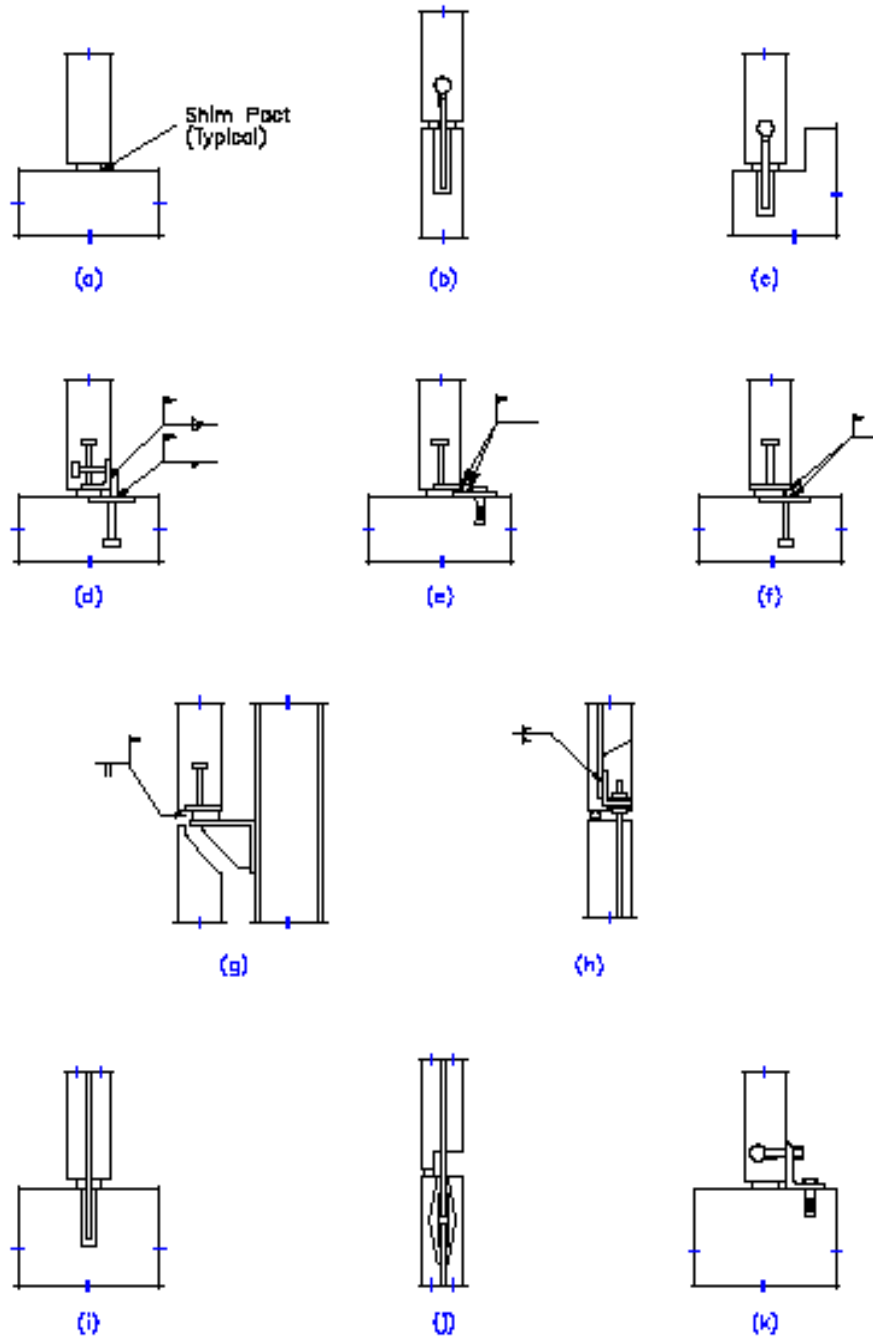
TYPICAL PANEL JOINT DETAILS



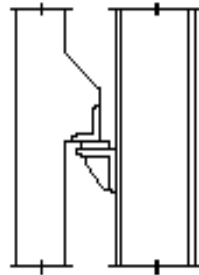
TYPICAL PILASTER DETAILS



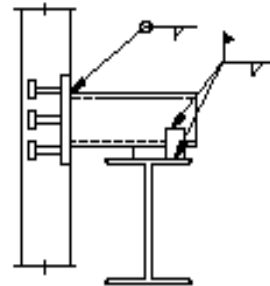
DIRECT BEARING CONNECTIONS



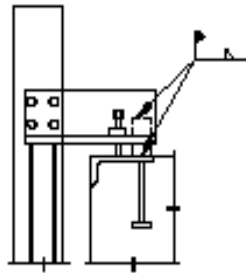
STEEL OR CONCRETE HAUNCH BEARING CONNECTIONS



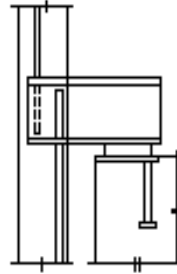
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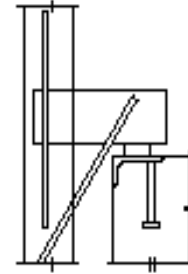
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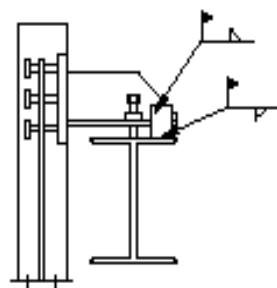
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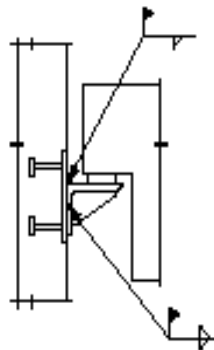
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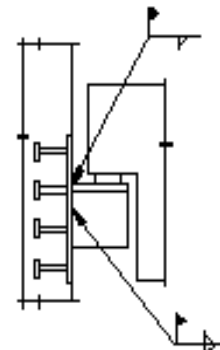
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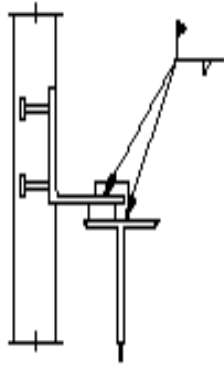


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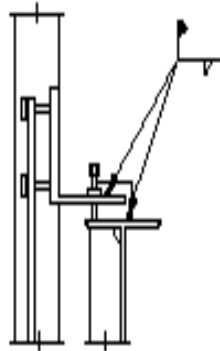
NOTE:

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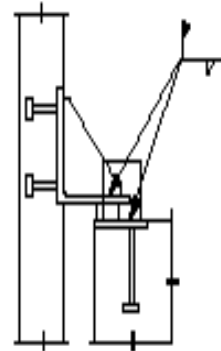
ANGLE SEAT BEARING CONNECTIONS



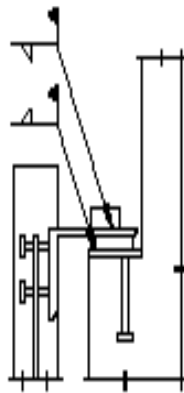
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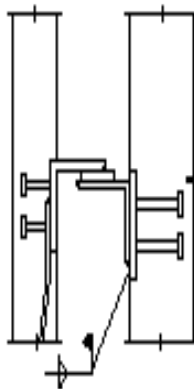
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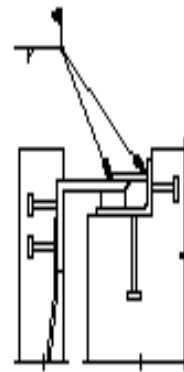
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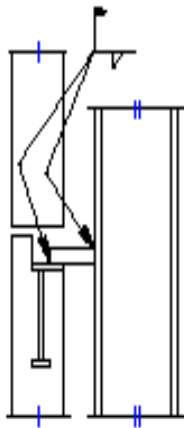


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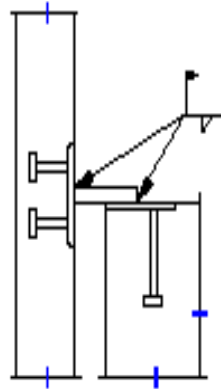
NOTE:

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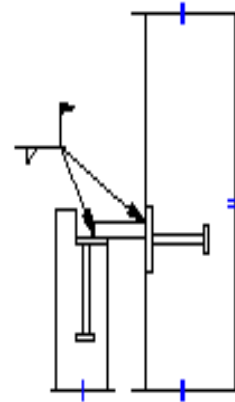
WELDED PLATE TIEBACK CONNECTIONS



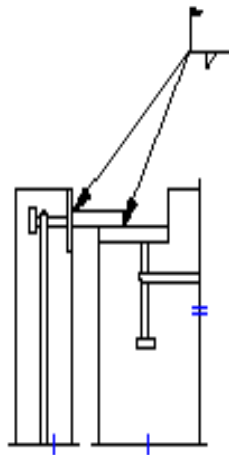
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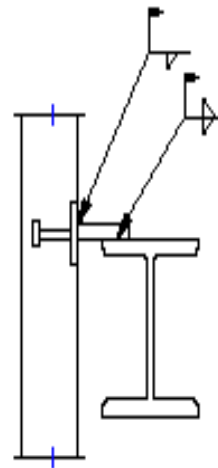
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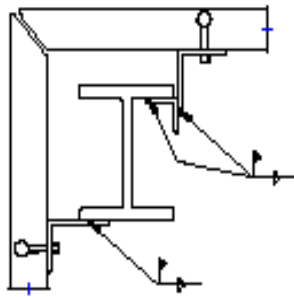
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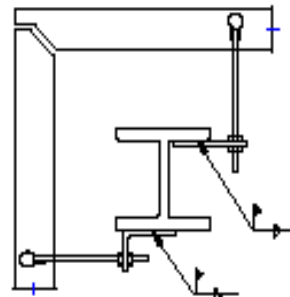
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TILT WALL CONNECTION DETAILS

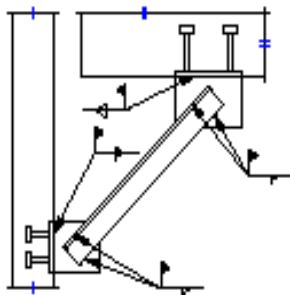
Welded Angle Tieback Connections



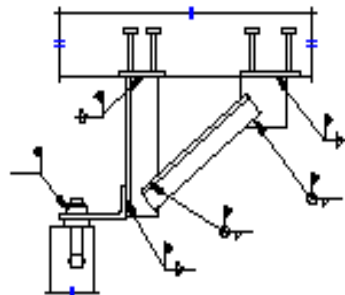
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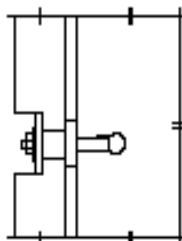


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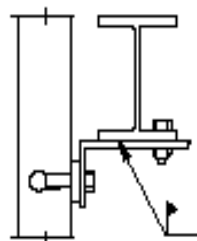


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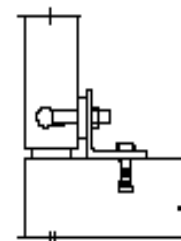
Bolted Tieback Connections



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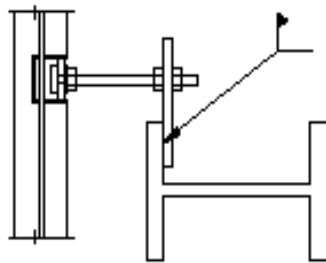
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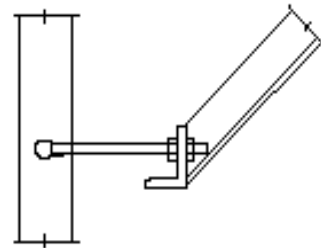
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TILT WALL CONNECTION DETAILS

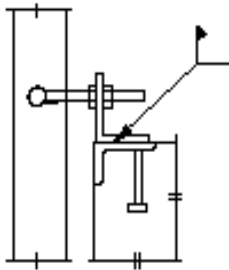
Threaded Rod Tieback Connections



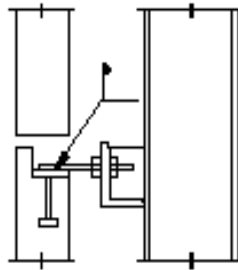
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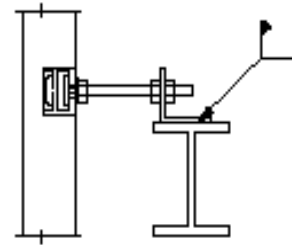
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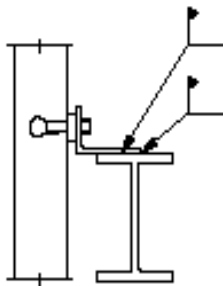


(d)

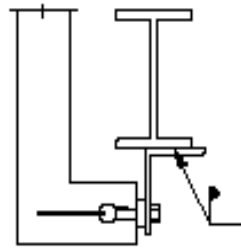


(e)

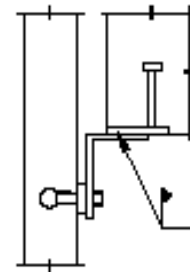
Bolted and Welded Tieback Connections



(a)



(b)

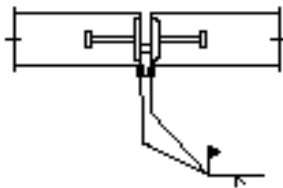


(c)

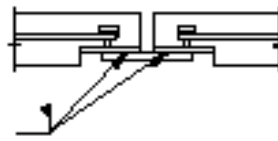


TILT WALL CONNECTION DETAILS

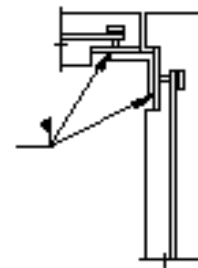
Welded Alignment Connections



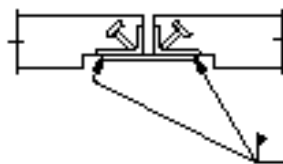
(a)



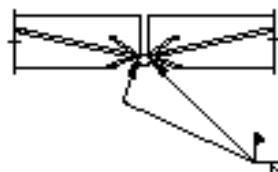
(b)



(c)



(d)

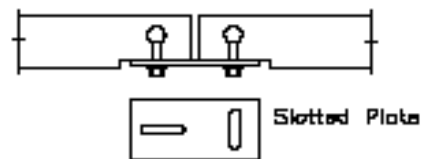


(e)

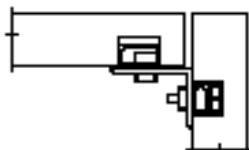
Bolted Alignment Connections



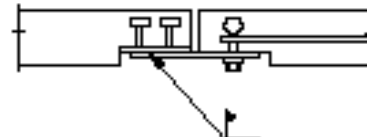
(a)



(b)



(c)



(d)

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SAMPLE DESIGN 1 : SOLID PANEL ON CONTINUOUS FOOTING

GIVEN:

$$\text{Thickness} = 7 \frac{1}{4}'' (91 \text{PSF})$$

$$f'_c = 3000 \text{ PSI}$$

$$f_y = 60 \text{ KSI}$$

$$P = 300 \text{ \#/ft}$$

$$e = 3'' + \frac{7.25}{2} = 6.63$$

$$\text{Wall Wt. @ } h/t = (91)(15.5) = 1411 \text{ \#/ft}$$

$$V = .3W = 27.3 \text{ \#/ft}$$

$$\text{Try \#5 @ 12} \quad (e_g = .00356)$$

$$L/t = 26(12) / 7.25 = 43 \rightarrow \text{SLENDER WALL DESIGN}$$

PROPERTIES

$$E_c = 57\sqrt{3000} = 3122 \text{ KSI}$$

$$n = \frac{E_s}{E_c} = 9.29$$

$$f_r = 5\sqrt{3000} = 274 \text{ PSI}$$

$$I_g = \frac{bt^3}{12} = \frac{(12)(7.25)^3}{12} = 381 \text{ in}^4$$

$$d = t/2 = 3.625''$$

$$S_g = \frac{bt^2}{6} = \frac{(12)(7.25)^2}{6} = 105.1 \text{ in}^3$$

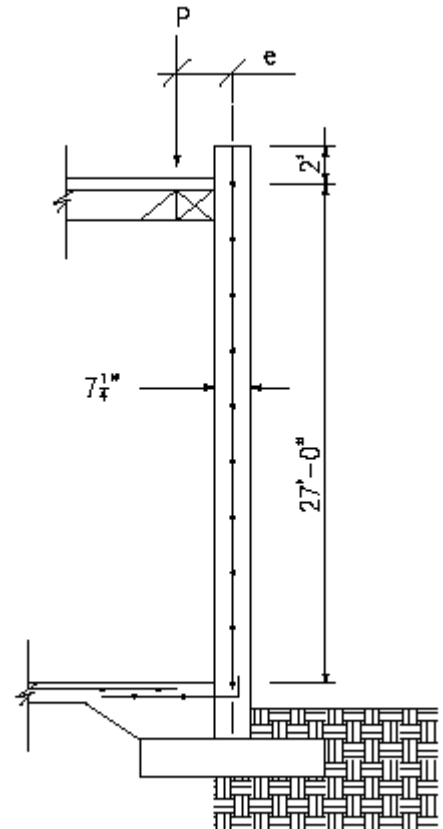
$$M_{CR} = S_g F_r = (105.1)(274) = 28,805 \text{ \#''}$$

I_{CR}:

$$P_U = 1.05(300 + 1411) = 315 + 1482 = 1797 \text{ \#/ft}$$

$$A_{se} = \frac{P_U + A_s f_y}{f_y} = \frac{1797 + (.31)(60,000)}{60,000} = .340 \text{ in}^2/\text{ft}$$

$$a = \frac{P_U + A_s f_y}{.85 f'_c b} = \frac{1797 + (.31)(60,000)}{.85(60,000)(12)} = .667''$$



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SAMPLE DESIGN 1 : SOLID PANEL ON CONTINUOUS FOOTING

$$c = \frac{a}{.85} = .785$$

$$I_{CR} = nA_{se}(d-c)^2 + \frac{bc^3}{3} = (9.29)(.340)(3.625-.785)^2 + \frac{(12)(.785)^3}{3} = 27.4 \text{ in}^4/\text{ft}$$

DEFLECTION:

$$M_s = Wl^2/8 = \frac{(27.3)(27)^2}{8} = 2.49 \text{ K}^-'/\text{FT} = 29.85 \text{ K}^-''/\text{FT}$$

$$M_n = A_{SE}(fy)(d-a/2) = (.340)(60)(3.625-.667/2) = 67.15 \text{ K}^-''/\text{FT}$$

$$\Delta_{CR} = 5(M_{CR}I^2)/48E_cI_g = \frac{5(28.81)(26 \times 12)^2}{48(3122)(381)} = .243''$$

$$\Delta_n = 5(M_nI^2)/48(E_cI_a) = \frac{5(67.15)(312)^2}{48(3122)(27.4)} = 7.96''$$

$$\Delta_s = \Delta_{CR} + \left(\frac{M_s - M_{cr}}{M_n - M_{cr}} \right) (\Delta_n - \Delta_{CR}) = .243'' + \left(\frac{29.85 - 28.8}{67.15 - 28.8} \right) (7.96 - .243) = .454'' = L/713 < L/150$$

REINFORCING:

$$W_U = 1.4(27.3) = 38.22 \#/\text{FT}$$

$$P_{OU} = 1.05(300) = 315 \#/\text{FT}$$

$$P_{DU} = 1.05(1411) = 1482 \#/\text{FT}$$

$$M_u = W_u I^2/8 + \frac{P_{DU}}{2} \Delta + \frac{P_{OU}}{2} \Delta + P_{OU^e} = \frac{(38.22)(27)^2(12)}{8} + \frac{(1482)(0)}{2} + (315)(0) + (315)(6.63) = 43.882 \text{ K}^-''/\text{FT} > M_{cr}$$

$$\therefore \Delta_1 = \Delta_{CR} + \left(\frac{M - M_{CR}}{M_s - M_{CR}} \right) \left[\frac{5(M_s - M_{CR})}{48EI_{CR}} I^2 \right] = .243 + \left(\frac{43.88 - 28.80}{29.85 - 28.80} \right) \left[\frac{5(29.85 - 28.8)}{48(3122)(2.4)} (312)^2 \right] = 2.03''$$

$$M_2 = 43.88 + \frac{(1428)(2.03)}{2} + (315)(2.03) = 46.02 \text{ K}^-''$$

$$\Delta_2 = .243 + \left(\frac{46.02 - 28.8}{29.85 - 28.8} \right) (.12) = 2.21'' \quad \dots = 1\% \text{ OK}$$

$$M_3 = 43.88 + \frac{(1482)(2.21)}{2} + (315)(2.21) = 46.21 \text{ K}^-''$$

$$\therefore M_U = 46.21 < fM_n = .9(67.5) = 60.4 \text{ K}^-''$$

OK

USE#5@12" C/C

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SAMPLE DESIGN 2: WALL PANEL W/OPENING

SEE PREVIOUS

EXAMPLE FOR DESIGN

DATA AND

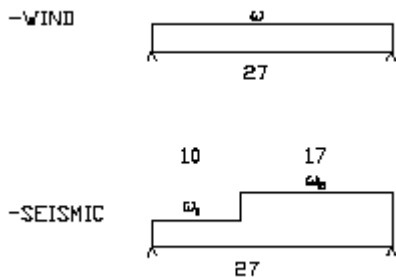
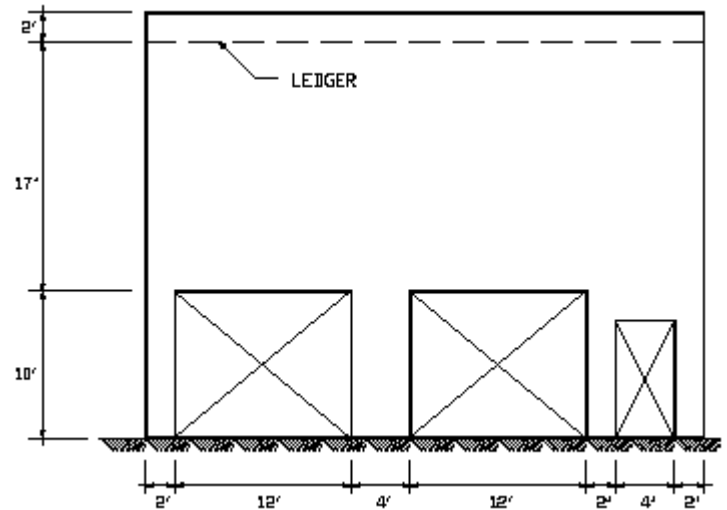
PARAMETERS

ALSO GIVEN:

Wind = 20 PSF

Doors span Horizontally

Light Weight



*DESIGN 4' STRIP ONLY

$$w = (20 \text{ PSF})(16') = 320 \text{ \#/FT}$$

$$M = \frac{(.32)(27)^2}{8} = 29.2 \text{ K'-'}$$

$$W_1 = 27.3(4') = 110 \text{ \#/FT}$$

$$W_2 = 27.3(4') = 110 \text{ \#/FT}$$

$$R_2 = \frac{1}{27} \left[\frac{437(27)^2}{2} - \frac{110(10)^2}{2} \right] = 5696 \text{ \#}$$

$$X = \frac{5696 \text{ \#}}{437 \text{ \#/FT}} = 13.03'$$

REINFORCING:

$$M = 5.69K(13.03') - \frac{(.422)(13.03)^2}{2} = 38.32 \text{ K'-'}$$

∴ SEISMIC GOVERNS

$$\text{--TRY } 48'' \times 7 \frac{1}{4} \text{ PIERW/6 - \#6 E.F. } (A_s = 2.64)$$

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SAMPLE DESIGN 2: WALL PANEL W/OPENING

$$d = 7\frac{1}{4} - 1\frac{1}{2} - \frac{3}{8} - \frac{3}{8} = 5"$$

$$e = .011$$

$$I_g = 48 \frac{(7.25)^3}{12} = 1524 \text{ in}^4$$

$$S_g = 48 \frac{(7.25)^2}{6} = 420.5 \text{ in}^3$$

$$M_{CR} = S f_r = (420.5)(274) = 115,217 \text{ #-in}$$

I_{CR}:

$$P_{OU} = 1.05(300 \text{ #/FT})(16') = 5,040 \text{ #}$$

$$P_{DU} = 1.05(91 \text{ PSF})(14')(16') = 21,403 \text{ #}$$

$$S = 26,443 \text{ #}$$

$$A_{se} = \frac{P_U + A_s f_Y}{f_Y} = \frac{26,443 + (2.64)(60,000)}{60,000} = 3.08 \text{ in}^2$$

$$a = \frac{P_U + A_s f_Y}{.85 f'_c b} = \frac{184,843 + (2.64)(60,000)}{.85(3000)(48)} = 1.510"$$

$$c = a / .85 = 1.777"$$

$$I_{CR} = 9.29(3.06)(5 - 1.777)^2 + \frac{(48)(1.777)^3}{3} = 385.1 \text{ in}^4$$

DEFLECTION: For simplification assume M_{max} and Δ_{max} occur @ MidSpan.

(Add 5% for Conserv.)



$$M_s = (38.32)(1.05) = (40.32)(12) = 482.8 \text{ K-}$$

$$M_n = (3.06)(60 \text{ KSI}) \left(5" - \frac{1.51}{2} \right) = 779.4 \text{ K-}$$

$$\Delta_{CF} = \frac{5(115.22)(27 \times 12)^2}{48(3122)(1524)} = .265"$$

$$\Delta_n = \frac{5(779.4)(324)^2}{48(3122)(385.1)} = 7.09"$$

$$\Delta_n = .265 + \left(\frac{482.8 - 115.2}{779.4 - 115.2} \right) (7.09 - .265) = 4.04" = L/80 > L/150 \quad \text{N.G.}$$

$$-- \text{TRY } 8\frac{1}{2}" \text{ THICK} \quad (106.3 \text{ PSF})$$



SAMPLE DESIGN 2: WALL PANEL W/OPENING

$$A_s = A_s' = 3.0 \text{ in}^2 (e = 0.010) \quad (\text{incl.comp.stl.in } I_{cr} \text{ calc.})$$

$$w_1 = (106.3)(0.3)(4') = 127.6 \text{ \#/FT}$$

$$w_2 = (106.3)(0.3)(16') = 510.2 \text{ \#/FT}$$

$$R_2 = \frac{1}{27} \left[\frac{(510.2)(27)^2}{2} - \frac{(127.6)(10)^2}{2} \right] = 6651 \#$$

$$X = \frac{6651}{510.2} = 13.03'$$

$$M = 6.651(13.03) - \frac{(510.2)(13.03)^2}{2} = 43.40 \text{ K-'}^2$$

$$M_s = (1.05)(45.4) = 45.6 \times 12 = 546.9 \text{ K-"}^2$$

$$d = 8\frac{1}{2} - 7\frac{1}{4} + 5 = 6.25 \quad d' = 2.25"$$

$$I_g = 48 \frac{(8.5)^3}{12} = 2456.5 \text{ in}^4$$

$$S_g = 48 \frac{(8.5)^2}{6} = 578 \text{ in}^3$$

$$M_s = (1.05)(45.4) = 45.6 \times 12 = 546.9 \text{ K-"}^2$$

$$M_{CR} = S f_r = 158.4$$

$$e = 3" + 8.5 / 2 = 7.25"$$

I_{CR} :

$$P_{OU} = 5,040 \#$$

$$P_{DU} = (1.05)(106.3)(14)(16) = 25,000 \#$$

$$S = \frac{25,000}{30,040 \#}$$

$$A_{se} = \frac{30.040 + 3.0(60)}{60} = 3.50 \text{ in}^2$$

$$a = \frac{210.0}{.85(3)(48)} = 1.72"$$

$$c = 1.72 / .85 = 2.02"$$

$$I_{CR} = (9.29)(3.5)(6.25 - 2.02)^2 + \frac{48(2.02)^3}{3} = 713.67 \text{ in}^4$$

DEFLECTIONS:

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SAMPLE DESIGN 2: WALL PANEL W/OPENING

$$M_n = (3.5)(60)(6.25 - \frac{1.72}{2}) = 1132^{K-}$$

$$\Delta_{cr} = \frac{5(158.4)(324)^2}{48(3122)(2456.5)} = .226"$$

$$\Delta_n = \frac{5(1132)(324)^2}{48(3122)(713.7)} = 5.56"$$

$$\Delta_s = .23" + \left(\frac{547 - 158.4}{1132 - 158.4} \right) (5.56 - .23) = 2.36" = l/138$$

Say OK w/Comp. Steel

REINFORCING:

$$M_U = 1.4(546.9) + \frac{P_o}{2} D + P_o D + \frac{P_o}{2} e = 765.7 + \frac{(25)(0)}{2} + 5.04(0) + \frac{(5.04)(7.25)}{2} = 784^{K-}$$

$$\Delta_1 = .226 + \left(\frac{784 - 158.4}{546.9 - 158.4} \right) \left(\frac{5(546.9 - 158.4)(324)^2}{48(3122)(713.67)} \right) = 3.29"$$

$$\therefore M_2 = 784 + \frac{(25)(3.29)}{2} + (5.04)(3.29) = 841^{K-}$$

$$\therefore \Delta_2 = .226 + \left(\frac{841 - 158.4}{546.9 - 158.4} \right) (1.907) = 3.58"$$

$$\therefore M_3 = 784 + \frac{(25)(3.58)}{2} + (5.04)(3.58) = 846.8^{K-}$$

$$M_U = 847^{K-} < FM_n = .9(1132) = 1019^{K-}$$

OK

USE 5 - #7 E.F.

CHECK AXIAL:

$$S P = 30K$$

$$f_a = \frac{P}{A} = \frac{30}{48(8.5)} = 74 \text{ PSI}$$

$$F_a = .04 f_e = .04(3000) = 120 > f_a$$

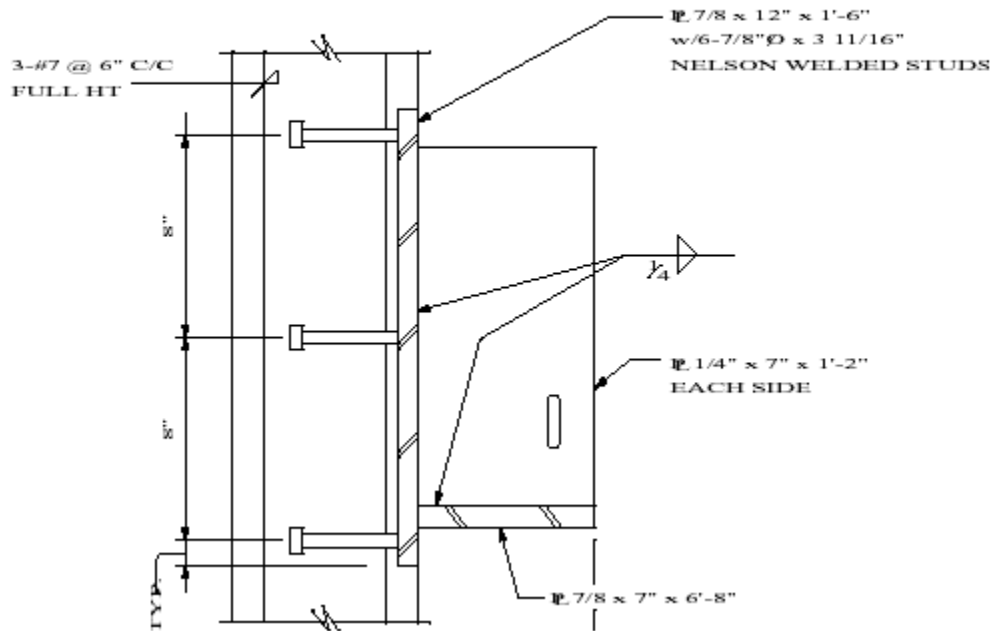
OK

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SAMPLE DESIGN 3: BEAM/PANEL CONNECTION

DESIGN CONN. TO RECEIVE $6\frac{3}{4}$ GLB

MAX GLB LOAD



$$c = 450 \text{ PSI} (1.25) = 562.5 \text{ PSI}$$

$$\text{Bearing } L = 7" - \frac{1}{2}" = 6\frac{1}{2}"$$

$$P_{\max} = 562.5(6.5)(6.25) = 24.7\text{K}$$

VERTICAL PLATE

$$V = 24.7\text{k}$$

$$M = \frac{27.4\text{K}(7")}{2} = 86.4\text{K}\cdot\text{"}$$

$$f_v = \frac{(24.7)(1.5)}{14(.25)(2)} = 5.3\text{KSI} < .4f_y = 14.4 \text{ KSI}$$

OK

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SAMPLE DESIGN 3: BEAM/PANEL CONNECTION

$$f_b = \frac{(86.4)(6)}{.25(14)^2(2)} = 5.3 \text{ KSI} < 21.6 \text{ KSI} \quad \text{OK}$$

USE Plate 1/4" x 7" x 1'2"

HORIZONTAL PLATE:

$$w = 24.7\text{K} / 6.75" = 3.66 \text{ K/in}$$

$$M = \frac{(3.66)(6.75)^2}{8} = 20.84^{\text{K-in}}$$

$$t = \sqrt{\frac{6(20.84)}{7"(27\text{KSI})}} = .813" \quad \rightarrow \frac{7}{8} \text{ Plate}$$

USE Plate 1/4" x 7" x 1'2"

STUDS:

$$V = 24.7\text{k}$$

$$V_{\text{STUD}} = 24.7\text{K} / 6 = 4.12\text{K}$$

$$T_{\text{STUD}} = \frac{86.4}{.85(2)(14)} = 3.63\text{K}$$

$$V_{\text{ALLOW}} = 27.19\text{K}$$

$$T_{\text{ALLOW}} = 11.63\text{K}$$

BENDING IN PANEL: $U \cong 1.55(D + L)$ (EQUIV.)

$$M_U = 1.55(24.7)(3.5" + \frac{7.25"}{2}) = 273^{\text{K-in}} = 22.7^{\text{K-ft}}$$

$$b = 12" + \frac{7}{8}" + 2(\frac{3}{8}) + 2(1\frac{1}{2}") = 16.6$$

$$d = 7\frac{1}{4}" - 1\frac{1}{2}" - 2(\frac{3}{8}") = 5"$$

$$F = \frac{bd^2}{12000} = \frac{(16.6)(5)^2}{12000} = .035$$

$$K_U = M_U / F = 22.7^{\text{K-ft}} / .035 = 656$$

$$(e = .0147)$$

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SAMPLE DESIGN 3: BEAM/PANEL CONNECTION

$$A_s = e b d = (.0147)(16.6)(5) = 1.22 \text{ in}^2$$

USE 3 - #7 Full HT

$$A_s = 1.8 \text{ in}^2$$

AXIAL:

Assume 4' Parapet

$$\text{Width} = 12" + 7 \frac{1}{4} (4) = 41"$$

$$A = 41(7.25) = 297 \text{ in}^2$$

$$P @ M_{\text{MAX}} = 24.7K + \frac{(12.5)(7.25)}{1000} \frac{(41)(4)}{12} = 25.94K$$

$$f_a = 26K / 297 = 87 \text{ PSI}$$

$$F_a = .04f_c' = .04(3000) = 120 \text{ PSI} < f_a$$

OK